

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical
Practical – X

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Class: TYIT	Batch:01
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a. Create a PWA With a Web App manifest file Including.

Code(manifest.json):-

```
{
  "name": "My PWA Application",
  "short_name": "MyPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "447x447",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "447x447",
      "type": "image/png"
    }
  ]
}
```

Code(index.html):-

```
<!DOCTYPE html>
<html>
<head>
  <title>My PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#0d6efd">
</head>
<body>
  <h1>My Progressive Web App</h1>
  <button onclick="showNotification()">
    Show Notification
  </button>
  <script src="app.js"></script>
</body>
</html>
```

Code(app.js):-

```
if ('serviceWorker' in navigator)
{
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker Registered'));
}
```

```
function showNotification() {
}
Notification.requestPermission().then(permission =>
{
  if (permission === 'granted')
  {
    navigator.serviceWorker.ready.then(registration =>
    {
      registration.showNotification('PWA Notification', {
        body: 'Push Notification Working Successfully!',
        icon: 'icons/icon1.png'
      });
    });
  }
});
```

Code(service-worker.js):-

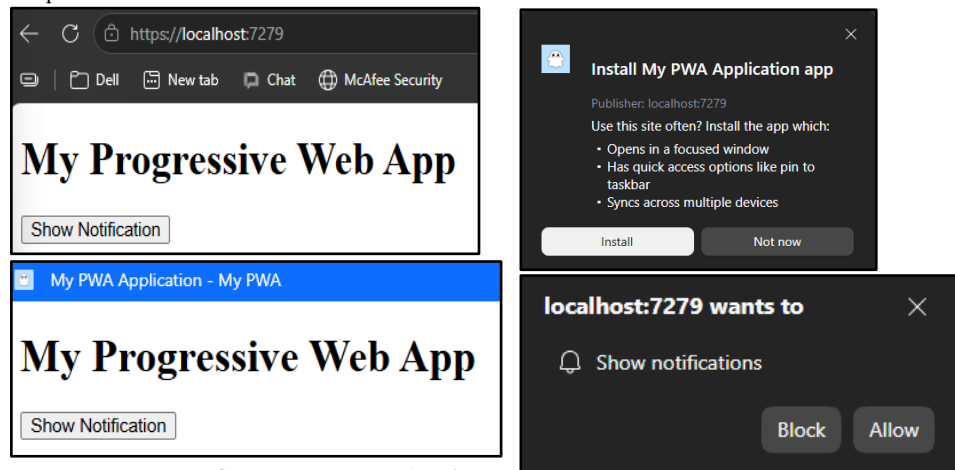
```
self.addEventListener('install', event =>
{
  console.log('Service Worker Installed');
});

self.addEventListener('activate', event =>
{
  console.log('Service Worker Activated');
});
```

Code(program.cs):-

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

Output:-



b. Develop an E-Commerce PWA with product listing and online cart functionality.

Code(program.cs):-

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

Code(index.html):-

```
<!DOCTYPE html>
<html>
```

```
<head>
  <title>E-Commerce PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#0466d6">
  <style>
    body {
      font-family: Arial;
      margin: 20px;
    }
    .container {
      display: flex;
      gap: 30px;
    }
    .cart-section {
      width: 30%;
      border: 1px solid gray;
      padding: 15px;
      min-height: 300px;
    }
    .product-section {
      width: 70%;
    }
    .products {
      display: flex;
      flex-wrap: wrap;
      gap: 15px;
    }
    .product {
      border: 1px solid gray;
      padding: 10px;
      width: 200px;
    }
    button {
      padding: 5px 10px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <h1>E-Commerce PWA</h1>
  <div class="container">
    <!-- Cart Section -->
    <div class="cart-section">
      <h2>Cart Items</h2>
      <ul id="cartList"></ul>
    </div>
    <!-- Product Section -->
    <div class="product-section">
      <h2>Products</h2>
      <div class="product">
        <h3>Laptop</h3>
        <p>Price: $ 5000</p>
        <button onclick="addToCart('Laptop')">
          Add to Cart
        </button>
      </div>
    </div>
  </div>
</body>
```

```

</div>
<div class="product">
  <h3>Mobile</h3>
  <p>Price: $ 2000</p>
  <button onclick="addToCart('Mobile')">
    Add to Cart
  </button>
</div>
<div class="product">
  <h3>Headphones</h3>
  <p>Price: $ 3000</p>
  <button onclick="addToCart('Headphones')">
    Add to Cart
  </button>
</div>
</div>
<script src="app.js"></script>
</body>
</html>

```

Code(app.js):-

```

if ('serviceWorker' in navigator)
{
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker Registered'));
}
let cart = JSON.parse(localStorage.getItem('cart')) || [];
displayCart();
function addToCart(product)
{
  cart.push(product);
  localStorage.setItem('cart', JSON.stringify(cart));
  displayCart();
}
function displayCart()
{
  let cartList = document.getElementById('cartList');
  cartList.innerHTML = "";
  cart.forEach(item => {
    let li = document.createElement('li');
    li.textContent = item;
    cartList.appendChild(li);
  });
}

```

Code(manifest.json):-

```

{
  "name": "E-Commerce PWA",
  "short_name": "ShopPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "447x447",

```

```

"type": "image/png"
},
{
  "src": "icons/icon2.png",
  "sizes": "447x447",
  "type": "image/png"
}
]
}
}
Code(service-worker.js):-
const CACHE_NAME = 'ecommerce-cache-v1';
const urlsToCache = [
  '/',
  '/index.html',
  '/app.js',
  '/manifest.json'
];
self.addEventListener('install', event => {
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(cache => {
        return cache.addAll(urlsToCache);
      })
  );
});
self.addEventListener('fetch', event => {
  event.respondWith(
    caches.match(event.request)
      .then(response => {
        return response || fetch(event.request);
      })
  );
});

```

Output:-

